

DCSS328

Deep Learning Application

Recent Trends in Deep Learning

What we will learn in this course

Lecture 1

Hanseob Kim

Fall 2026

Korea University

Computer Science and Software Engineering

Today's Core Question

“Why can AI solve an Olympiad Problems, but still struggle to clean a Room”



- Relatively simple/clear rule
- Clear Answer



- Noise
- Physics rule
- Long planning & Latency
- Safety

NVIDIA GTC Taipei 2026 Keynote

<https://www.youtube.com/watch?v=wSp6AiNIrsY>

2012 ALEXNET



1

PERCEPTION AI

SPEECH RECOGNITION
DEEP RECSYS
MEDICAL IMAGING

2

GENERATIVE AI

DIGITAL MARKETING
CONTENT CREATION

3

AGENTIC AI

CODING ASSISTANT
CUSTOMER SERVICE
PATIENT CARE

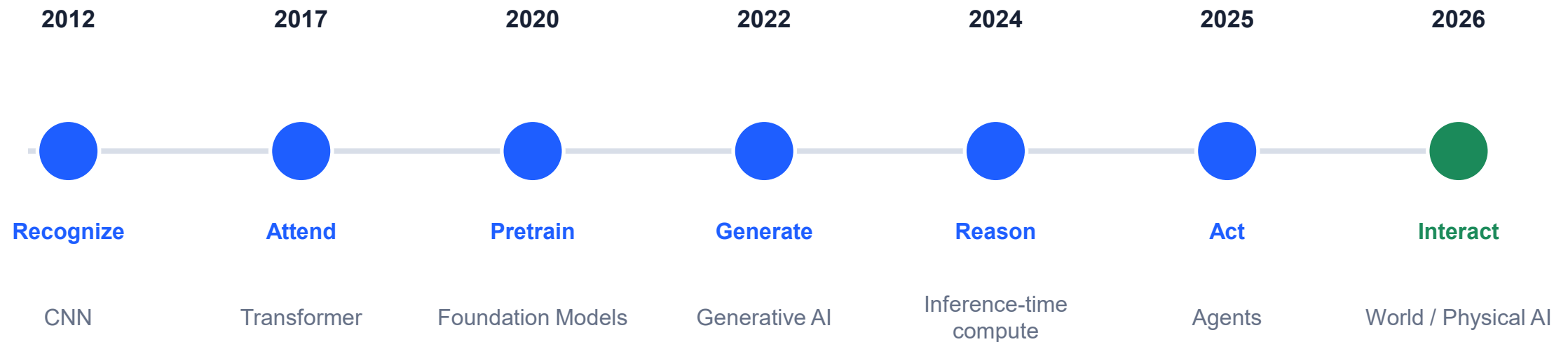
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PHYSICAL AI

AUTONOMOUS VEHICLES
GENERAL ROBOTICS

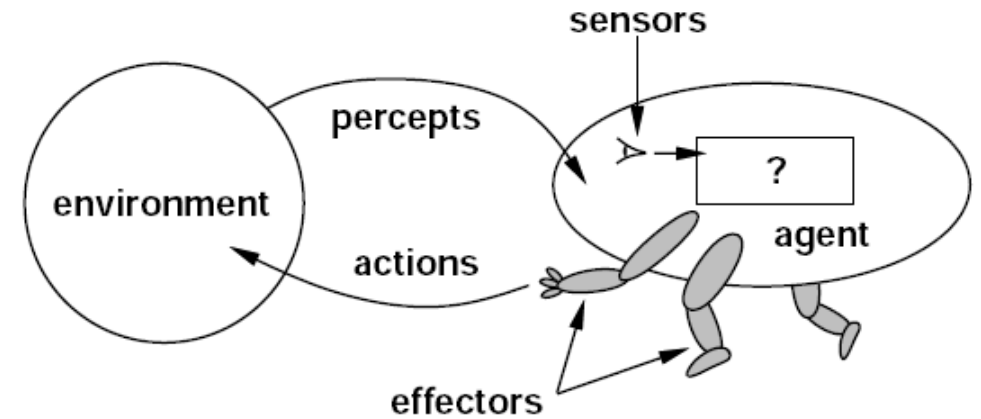


The AI Trends and its Capability Change



Model capability **expanded from pattern recognition to systems that reason, use tools, and interact with environments.**

The AI Trends and its Capability Change



Key five words

PERCEIVE

images · audio · 3D

GENERATE

text · image · video · motion

REASON

search · verify · revise

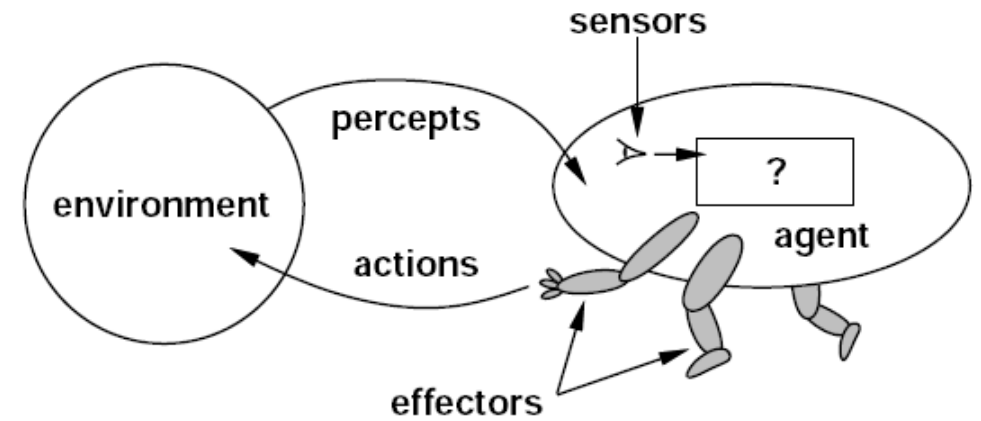
ACT

tools · APIs · computers

INTERACT

worlds · avatars · robots

The AI Trends and its Capability Change

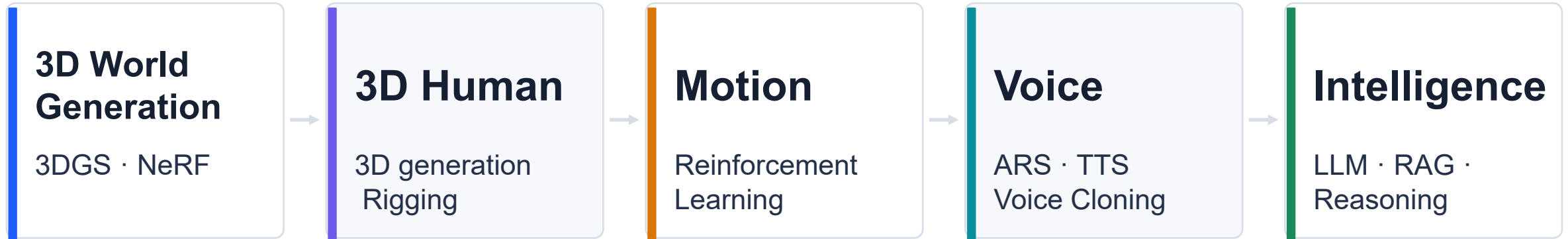


Prediction is not disappeared; still underneath everything

The difference is that modern systems wrap prediction with reasoning, tools, memory, simulation, and feedback.

From AI trends to our course

Our course focuses on the components needed to build an interactive AI experience in XR



- 01** **Foundation Models & Efficient AI**
- 02** **Reasoning & Test-Time Scaling**
- 03** **Multimodal & Generative AI**
- 04** **Agentic AI**
- 05** **Building Interactive Intelligence**
- 06** **Digital Humans & Embodied Motion**
- 07** **Speech & Voice**
- 08** **World Models & Physical AI**

01

Foundation Models & Efficient AI

From “train **One Model** per **SPECIFIC task**” to “**Adapt** and compose pretrained capabilities.”

Deep Learning Application in 2026

The Old Pattern vs. the New Pattern

TRADITIONAL DEEP LEARNING

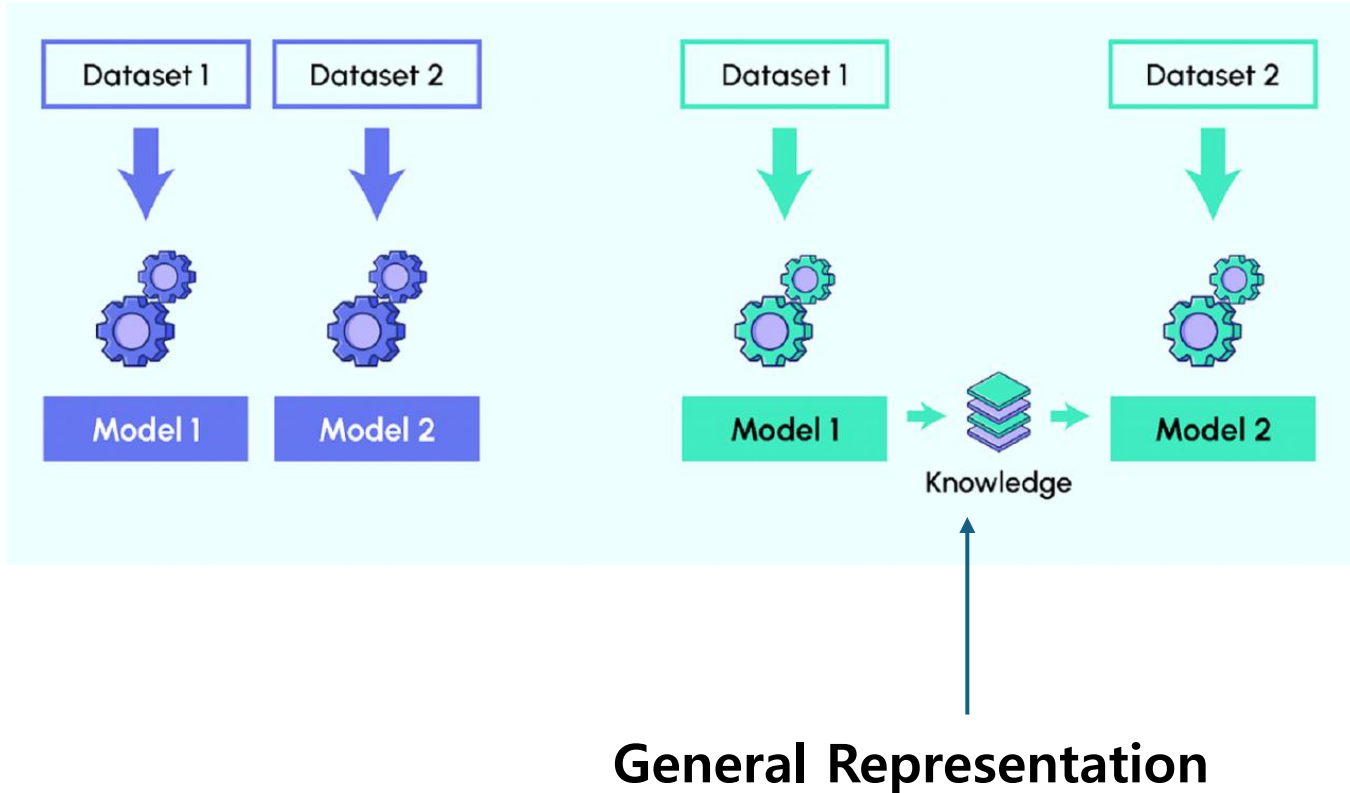
Dataset A → train Model A → Task A

Dataset B → train Model B → Task B

Dataset C → train Model C → Task C

Strong specialization, repeated training cost.

The Old Pattern vs. the New Pattern



MODERN FOUNDATION MODEL

Large-scale pretraining



General representations or capabilities



Prompting / retrieval / **Fine-Tuning**



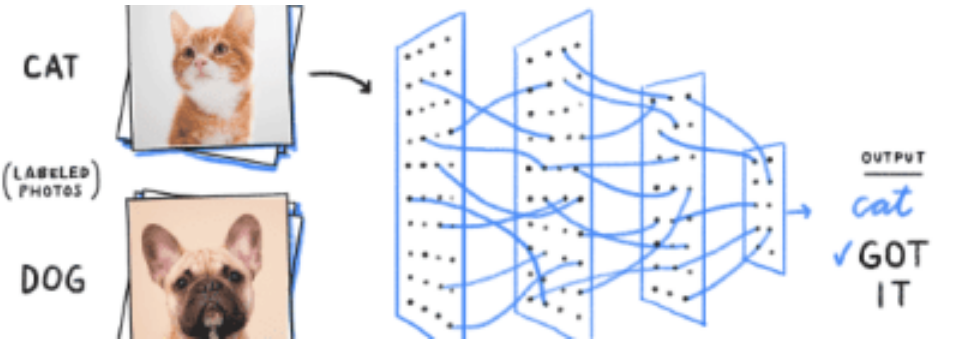
Many downstream tasks

The model becomes a **reusable** capability layer

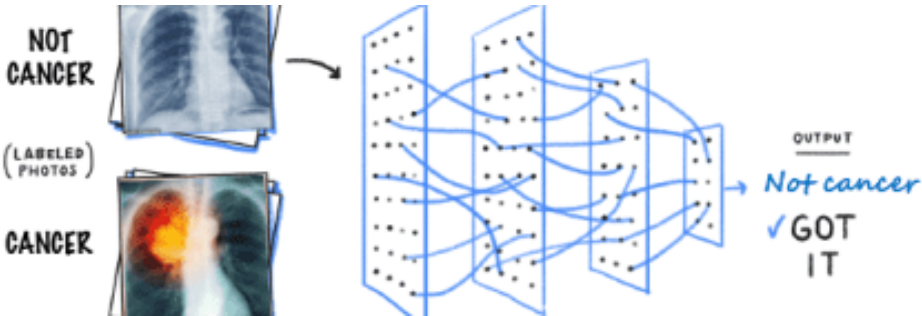
REUSE

The Old Pattern vs. the New Pattern

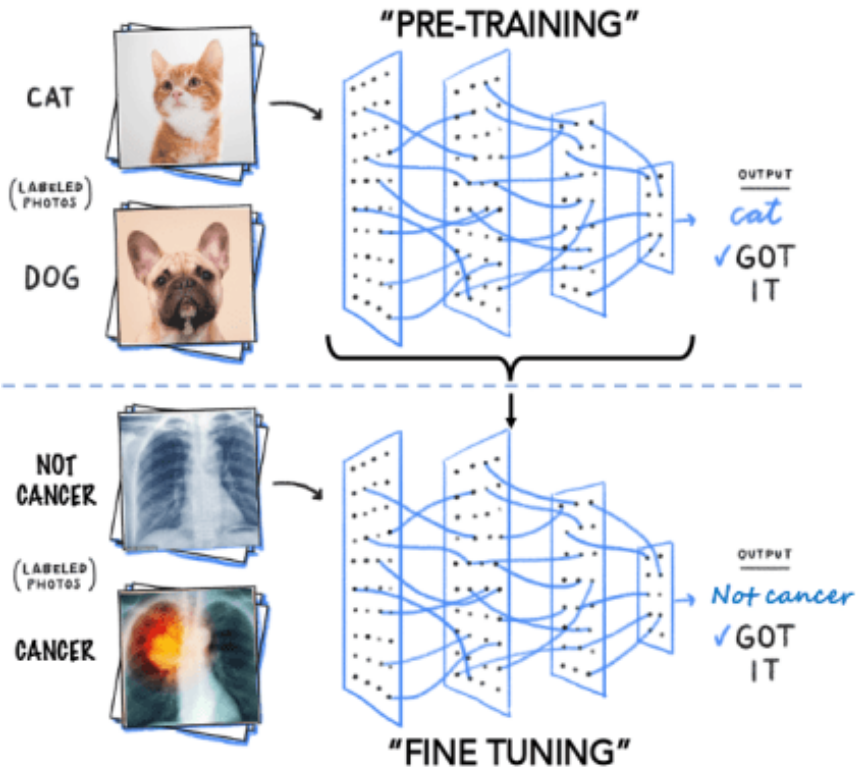
TRADITIONAL DEEP LEARNING



Strong specialization, repeated training cost



MODERN FOUNDATION MODEL

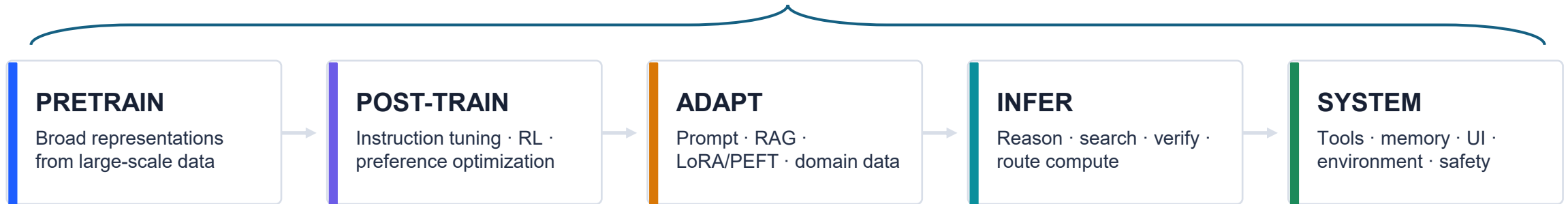


reusable trained weights

AI Modern AI Stack

Progress happens at multiple layers — not only in model architecture.

AI Model can be organized into Five Layers



Modern AI engineering often means selecting where to **inject** task-specific information and where to spend compute

In the past the focus was on how well the model itself was trained,

Now the important **AI engineering decision is where to allocate resources** among pre-training, post-training, RAG/context, inference-time compute, and agent/system.

Scaling Has More Than One Axis

Intelligence gains increasingly depend on how computation is allocated

Past

More Large Model + More Large Data + More Time for Train → GOOD AI

Model Scale

Parameters
architecture
Mixture-of-Experts

Makes the brain
itself bigger

Data Scale

Quantity
quality
deduplication
synthetic data

Trains with
more/better materials

Training Compute

Pretraining
post-training
reinforcement learning

Trains for longer
and better

Inference Compute

Sampling
search
verification
tool use

You think longer when
you receive a problem.

A model does not have to spend the same amount of computation on every query

- Simple question – answer quickly
- Difficult question – more time reasoning, searching or verifying

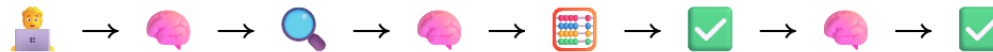
Scaling Has More Than One Axis

Easy Question



Low compute / Shorter Inference

Hard Question



High compute / Longer Inference

Key Question

How much compute should we allocate to each query ?

When Capability Converges, System Trade-offs Matter

Question: Which model should we use?

QUALITY

Does it solve the task?

COST

How much compute per request?

LATENCY

Can it respond in time?

RELIABILITY

Does it succeed repeatedly?

TOOL USE

Can it operate in a workflow?

“Best model” is becoming a task- and system-dependent choice.

Model A

Accuracy : 95 %
Response Time : 20 s
Fee : \$0.2
Usability : Good

vs

Model B

Accuracy : 93 %
Response Time : 1 s
Fee : \$0.01
Usability : So So

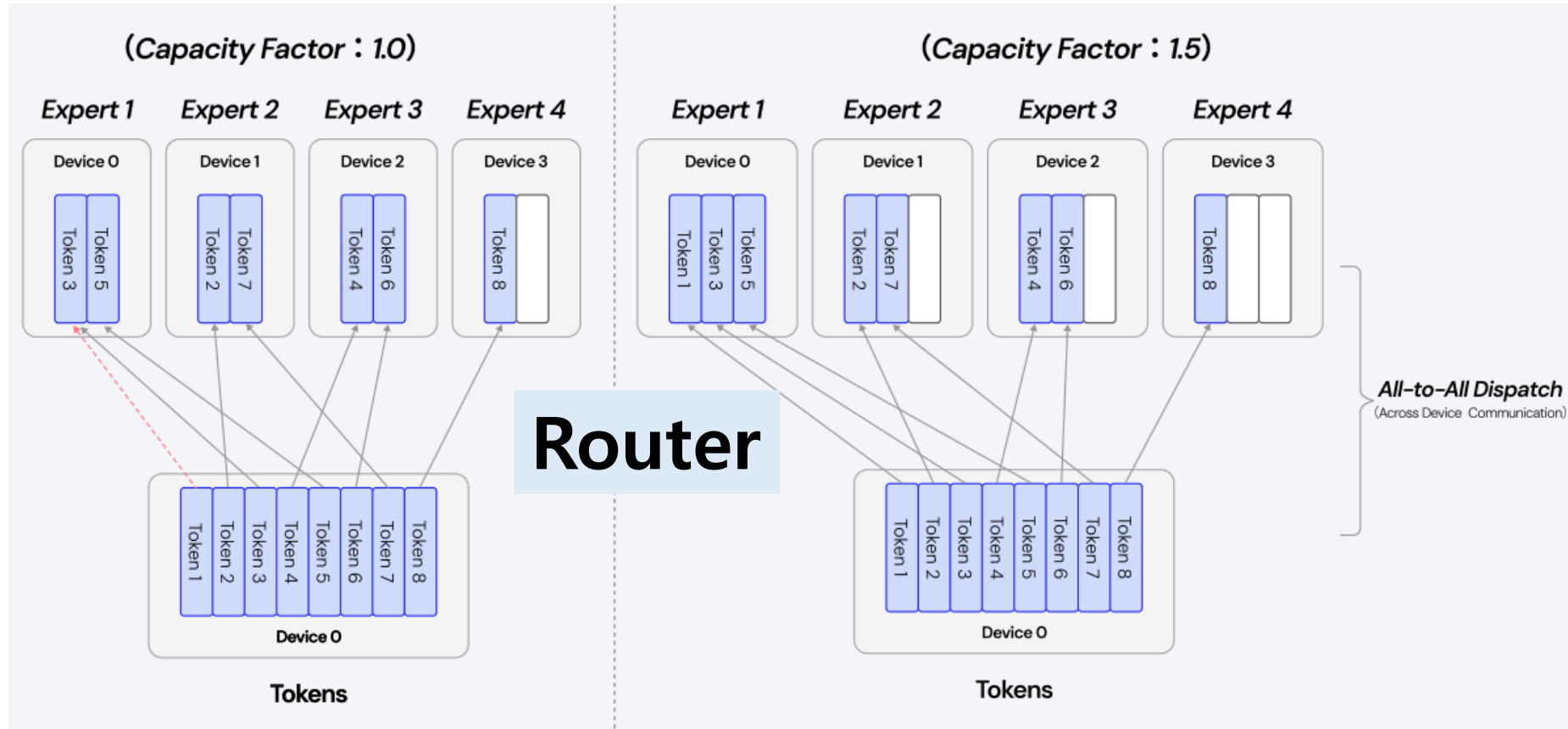
Scaling AI

+

How to make large models efficient ?

Efficiency

Sparse Models and Mixture-of-Expert (MoE)



Why it matters

- Higher capacity at lower active compute
 - Useful for cloud, edge, XR, and robotics constraints
 - Better memory / throughput trade-offs
- Total parameters $\neq$ active parameters**

How do we adapt a foundation model to a specific task?

PROMPTING

changes the instructions without changing the model weights

RAG

adds external knowledge at inference time

LoRA & PEFT

updates only a small part of the model

QUANTIZATION

reduces memory and computation

DISTILLATION

transfers capabilities to a smaller model

ROUTING

decides which model or how much compute to use

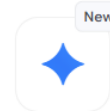
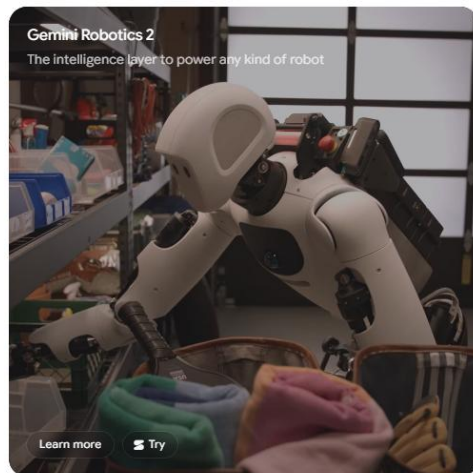
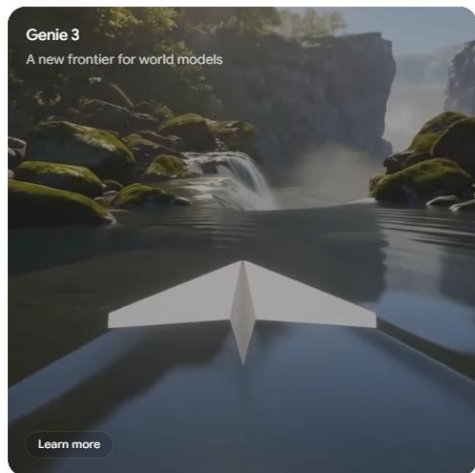
- They are just toolbox
- We choose where to inject task-specific information based on the problem and system requirements

Recent Trends

Native Multimodal Intelligence

Google DeepMind · Current Gemini / Omni model family (2026)

Text, image, audio, video, code, tools
inside one reasoning and action layer



Gemini

Frontier intelligence with action

TRY IT IN

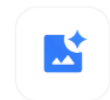
[Gemini app](#) >

[Google AI Studio](#) >

[Google Antigravity](#) >

[Gemini Enterprise Agent Platform](#) >

[Learn more](#)



Nano Banana 🍌

Create and edit images with Gemini Image

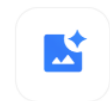
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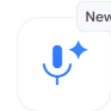
Imagen

Our leading text-to-image model

TRY IT IN

[Google AI Studio](#) >

[Gemini Enterprise Agent Platform](#) >



Gemini Audio

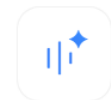
Advanced real-time audio models, built on Gemini

TRY IT IN

[Google AI Studio](#) >

[Gemini Enterprise Agent Platform](#) >

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Lyria

Our most advanced music generation model

TRY IT IN

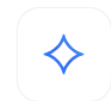
[Gemini app](#) >

[Google Flow Music](#) >

[Google AI Studio](#) >

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Gemma

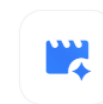
Our most capable open models

BUILD WITH GEMMA

[Google AI Studio](#) >

[Run Gemma](#) >

[Developer docs](#) >



Gemini Omni

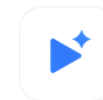
Create anything from anything, starting with video

TRY IT IN

[Gemini app](#) >

[Google Flow](#) >

[Learn more](#)



Veo

Our leading video generation model

TRY IT IN

[Gemini app](#) >

[Google Flow](#) >

[Google AI Studio](#) >

[Gemini Enterprise Agent Platform](#) >

[Learn more](#)



Gemini Embedding

State-of-the-art multimodal embedding model

BUILD WITH GEMINI EMBEDDING

[Gemini API docs](#) >

[Google Cloud docs](#) >

<https://deepmind.google/models/>

02

Reasoning & Test-Time Scaling

The model **DOSE NOT** have to spend the same amount of computation on every problem.

Deep Learning Application in 2026

Why Reasoning Became a Separate Capability

FAST RESPONSE

Simple Question



One forward generation



Answer

Great for routine tasks, but limited opportunity to explore alternatives or verify mistakes

DELIBERATE RESPONSE

Difficult Question



Propose → test → revise → verify



Answer

Spend extra inference compute when difficulty or uncertainty is high.

Reasoning can be viewed as an Inference Strategy:

Spending more compute when the problem requires it

Train-Time Scaling vs. Test-Time Scaling

TRAIN-TIME Scaling

More pretraining / RL / post-training



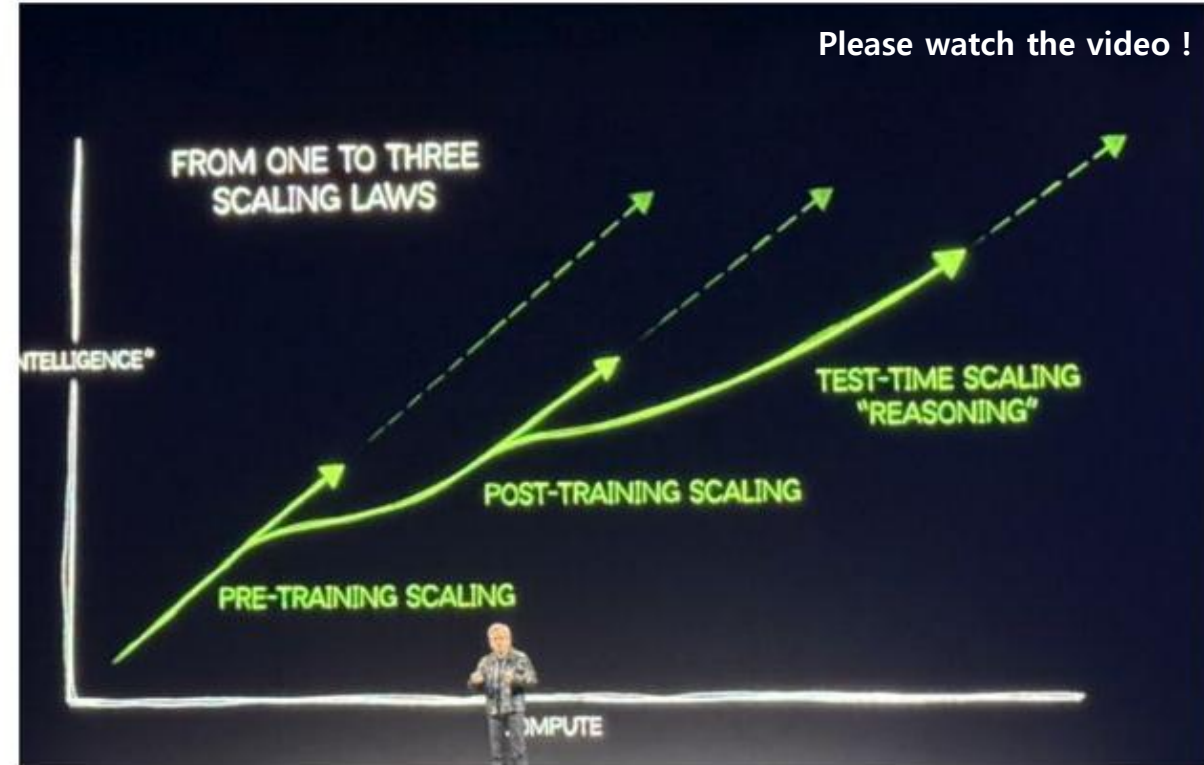
A better policy before deployment

TEST-TIME Scaling

More search / sampling / verification / tool use



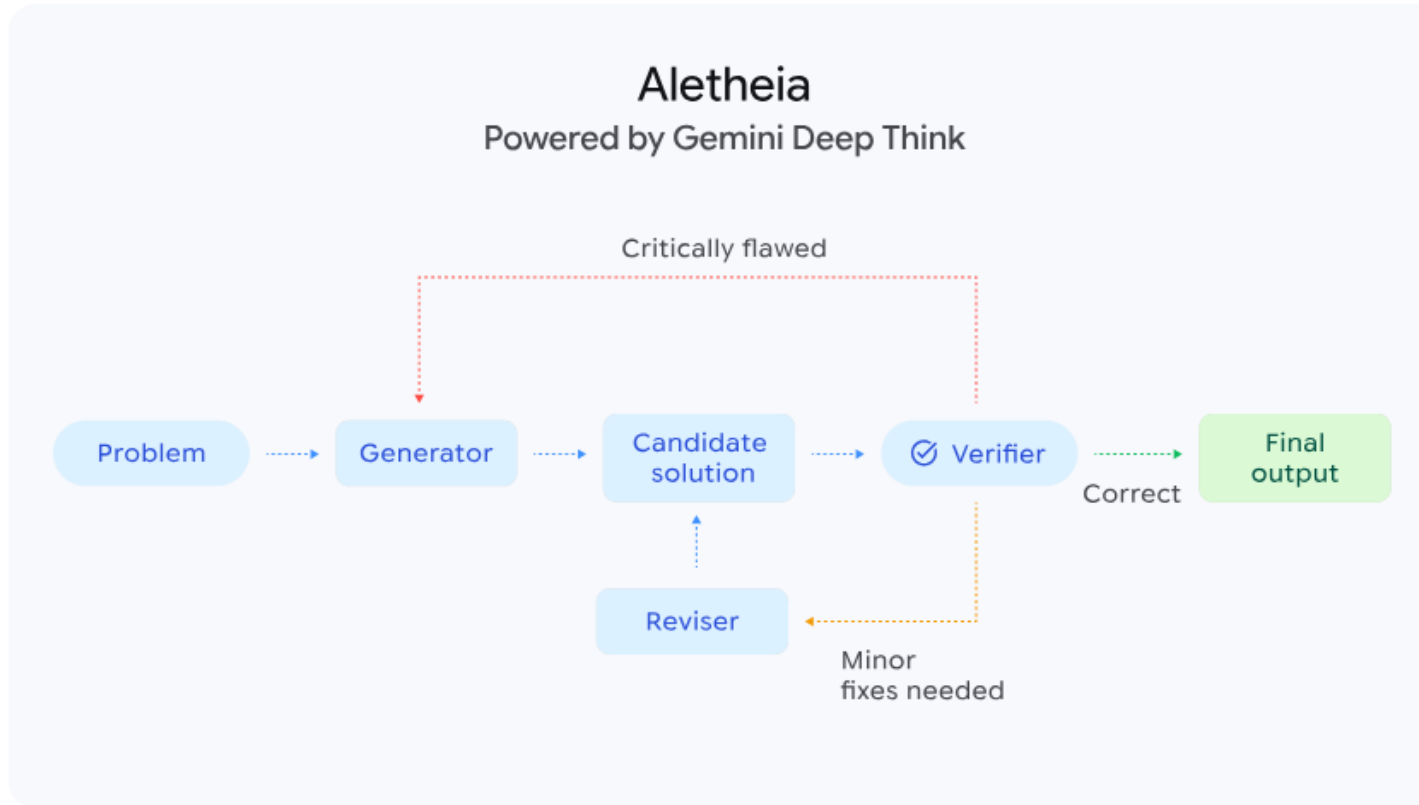
Potentially a better answer for this specific query



Key question:

How much computation should this specific query receive?

Reasoning Can Be a System Loop



Overview of Aletheia, a math research agent powered by Deep Think that can iteratively generate, verify, and revise for research-level math problems.

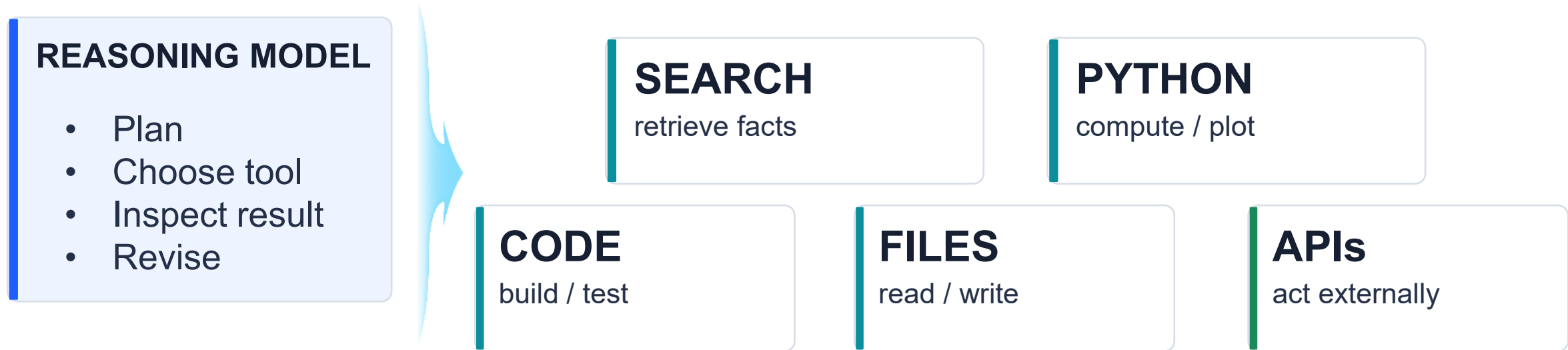
<https://deepmind.google/blog/accelerating-mathematical-and-scientific-discovery-with-gemini-deep-think/>

The “reasoner” is increasingly an orchestration of generation, checking, tools, and selective retries

Reasoning + Tools

Example loop: search → compute → inspect → revise → answer

A model **can pivot based on external evidence** instead of guessing through missing information.



The important capability is knowing when to use which tool and **then incorporating the tool result** into the next reasoning step

Benchmarks Are Moving — and Breaking

State of the art is conditional on the benchmark and **Evaluation Protocol.**

HARDER BENCHMARKS

Frontier performance on Humanity's Last Exam improved by roughly 30 percentage points in one year.

BENCHMARK SATURATION

Once an evaluation saturates, it stops telling us where systems fail.

BENCHMARK QUALITY

A review highlighted invalid -question rates from 2% to 42% across widely used evaluations.

SOTA should not be defined by just one number

Jagged Intelligence

The same system can be extraordinary on one task and surprisingly weak on another.

❖ VERY HARD

On ClockBench, the top model read analog clocks correctly 50.6% of the time vs. 90.1% for humans.

❖ SEEMINGLY SIMPLE

Gemini Deep Think achieved gold-medal-level performance at the 2025 International Mathematical Olympiad.

Do not infer general competence from a single spectacular demo

Limits of Reasoning

- 1 Wrong verification** A confident checker can validate a flawed path.
- 2 Excess compute** More tokens can add latency / cost without better quality.
- 3 Tool error** A correct plan can fail because browsing, code, or APIs fail.
- 4 Benchmark issues** Leakage, ambiguity, and invalid questions distort comparisons.
- 5 Monitorability** Internal reasoning traces are useful signals, but safety and control remain active research topics.

03

Multimodal & Generative AI

Text is becoming one modality among many, e.g., audio, image, video, code, 3D data, action

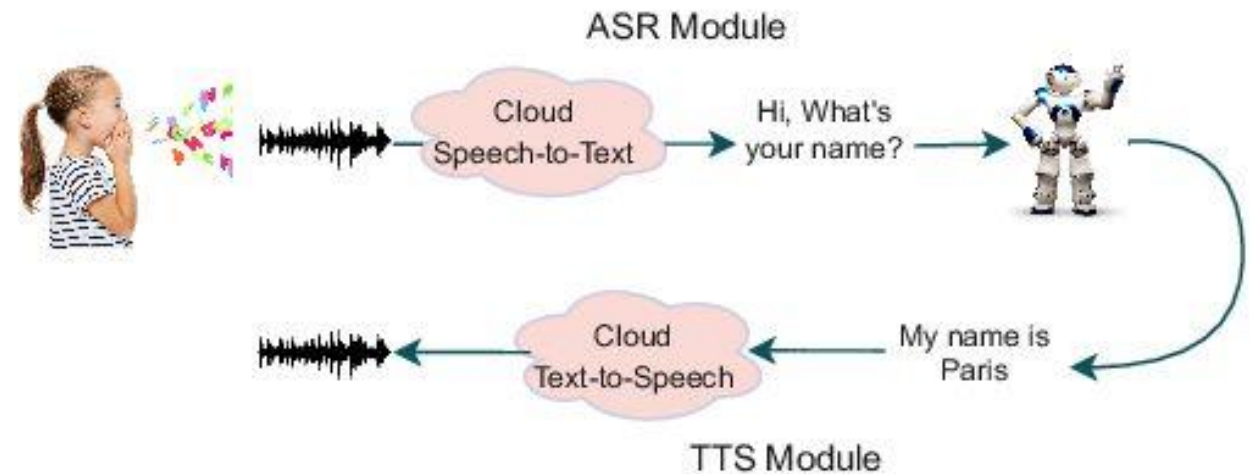
And generation is moving toward dynamic, interactive worlds.

Deep Learning Application in 2026

From Modality Pipelines to Native Multimodality

In past, Separate Module used

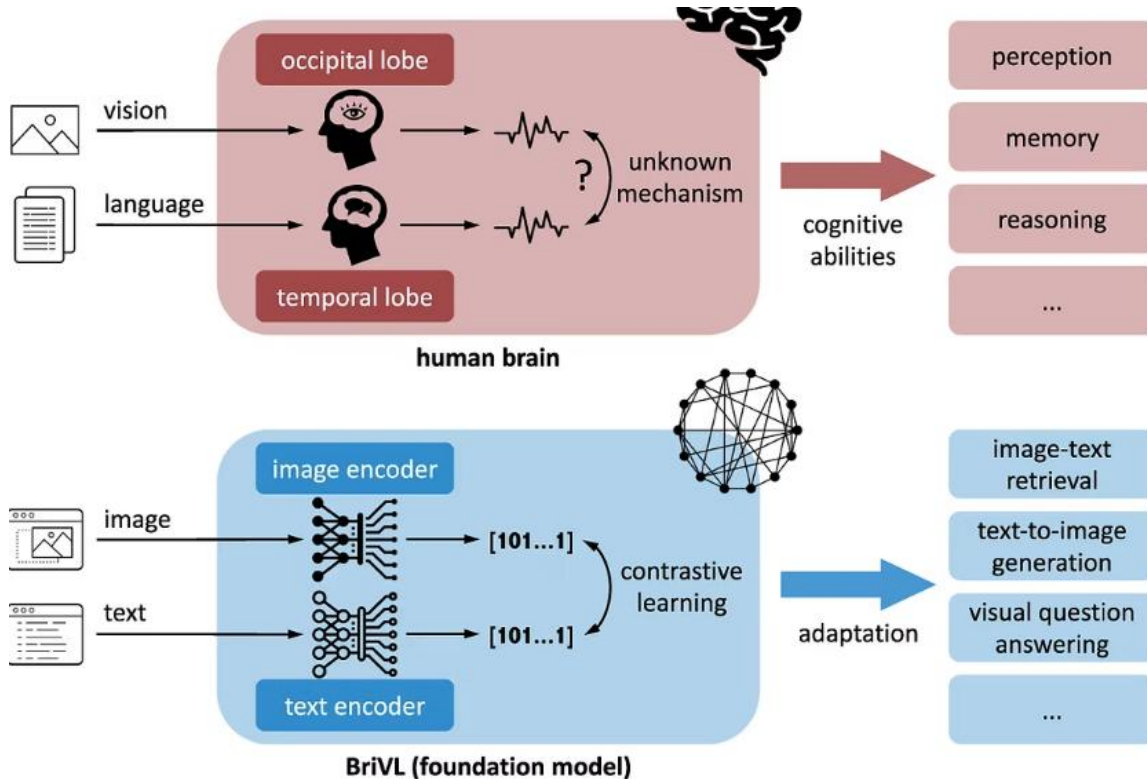
- Image encoder → text model
- Speech: ASR → LLM → TTS
- Video model → captioner
- Each modality often had a specialized interface.



Latency / Error could be introduced when converting Data to Text

MULTIMODAL

From Modality Pipelines to Native Multimodality



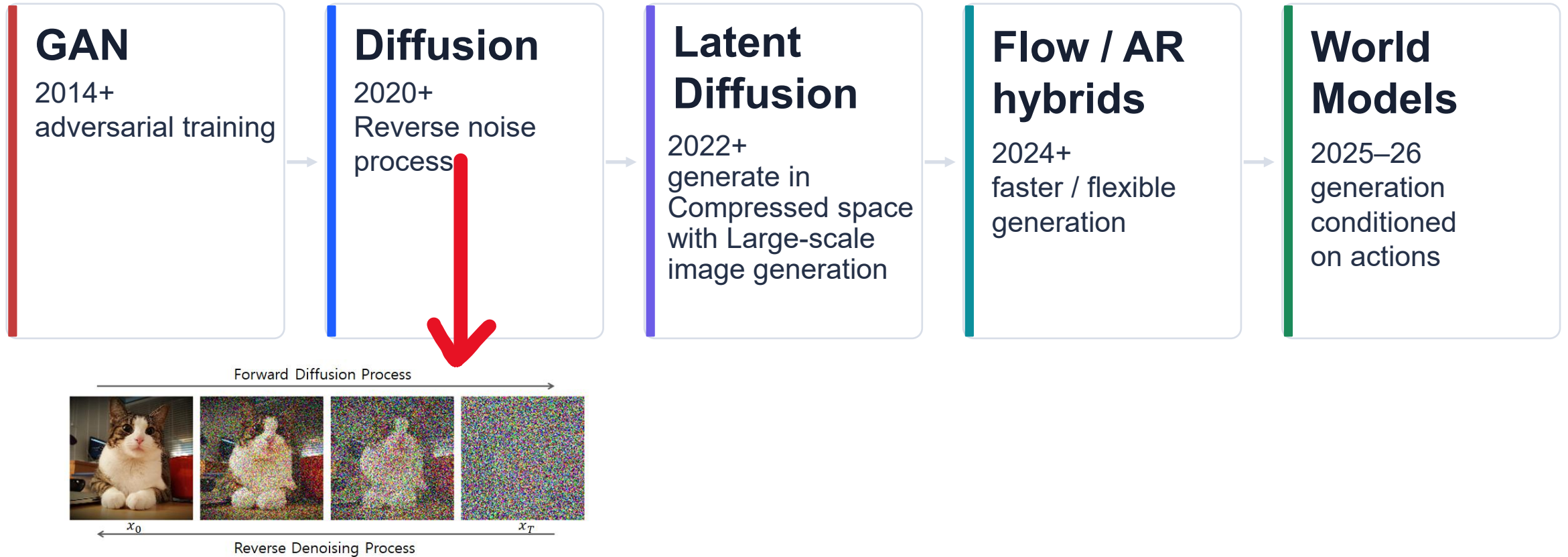
NATIVE MULTIMODAL ERA

- Text + image + audio + video + code
- ↓
- Shared reasoning / tool-use layer
- ↓
- Multimodal output or action
- Modality switching becomes part of one inter action loop.

The important advantage is that the **system does not always need to change everything to text first.**

History Generative Modeling

<https://www.youtube.com/watch?v=IxHXQpk5kkg>



Trajectory: static samples → coherent/consistent sequences → action-conditioned environments

Video Generation Is Becoming a World-Learning Problem

Evaluation increasingly asks whether generated video reflects world behavior — not only whether it looks real.

VISUAL FIDELITY

- Sharp Images
- temporal consistency
- cinematic quality

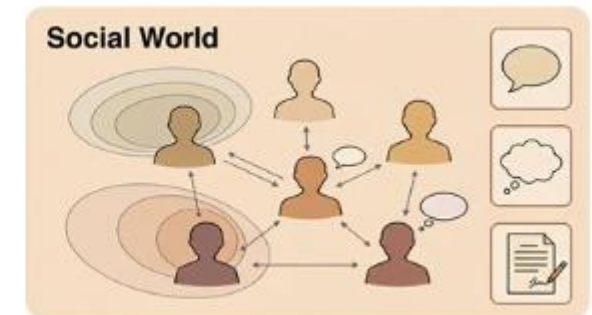
SEMANTIC CONTROL

Text / image steering
object control
camera / event control

PHYSICAL BEHAVIOR

Buoyancy
collision-like interactions
navigation / persistence

Visual realism and world correctness are not the same thing



Will AI understand Dynamics in physical World and Social World??

<https://deepmind.google/models/veo/>

Promptable Perception: SAM(segment anything model)

A specialist foundation model can become a perception tool inside a larger agentic system.

TEXT

"striped red umbrella"

POINT / BOX

interactive visual prompt

EXEMPLAR

show one example object

<https://www.aidemos.meta.com/segment-anything>



Isolate any part of an image or video instantly

Just type a phrase, and the model instantly finds every matching object across your media.



Turn your segments into dynamic, editable media

Experiment with new possibilities through one-click effects like blurring, cloning, and more.



Create reusable templates that speed you up

Advanced media tools for all. Save, download, and share with the community.

SAM 3 / 3.1

Detect → segment → track

2026 DIRECTION

- Unified detection + segmentation + tracking
- Text / exemplar / visual prompts
- Public playground
- Specialist vision tools become callable components.

04

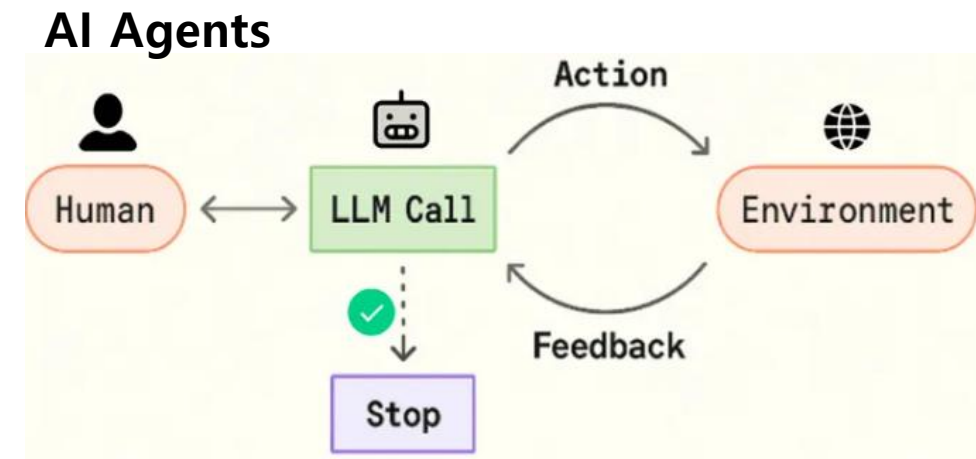
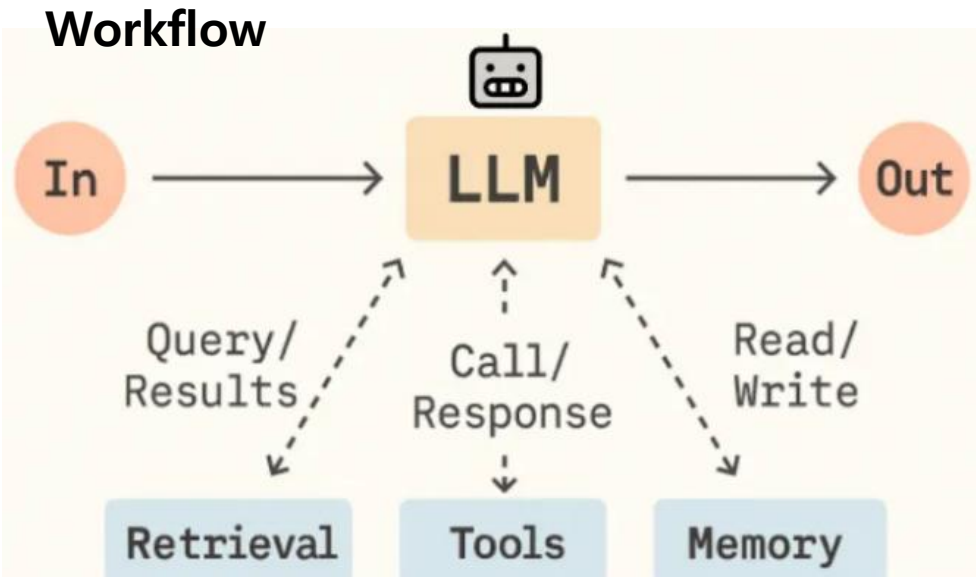
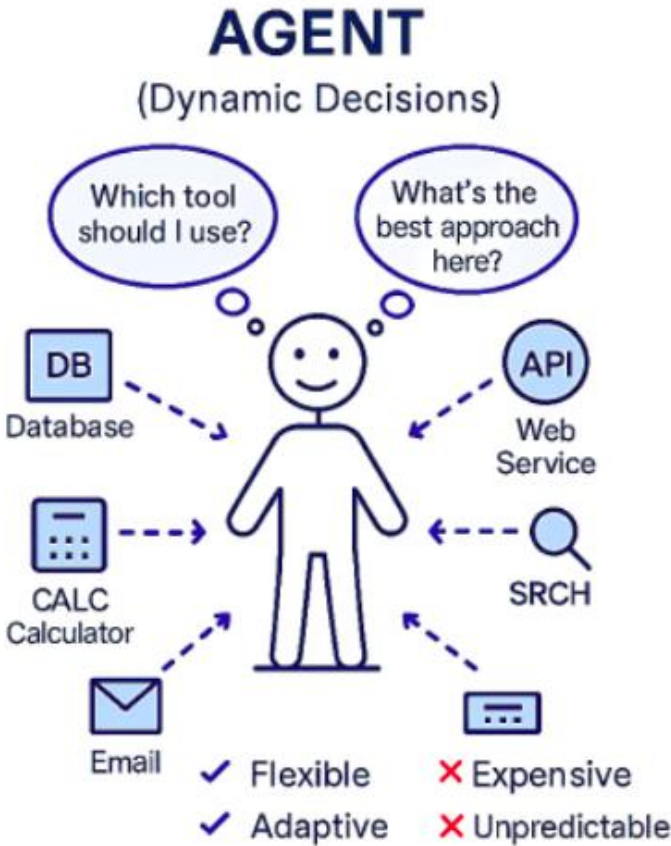
Agentic AI

When a model can select tools, observe results, and keep going, it becomes part of an acting system.

Output is no longer just text; Output may change an external environment!

Deep Learning Application in 2026

Workflow vs. Agent



<https://ai.plainenglish.io/ai-workflows-vs-ai-agents-vs-multi-agentic-systems-a-comprehensive-guide-f945d5e2e991>

AGENTS

A Minimal/Basic Agent Loop

Learn / Reflect

Improve performance
based on **feedback**

Sense / Observe

Collect inputs from
the environment

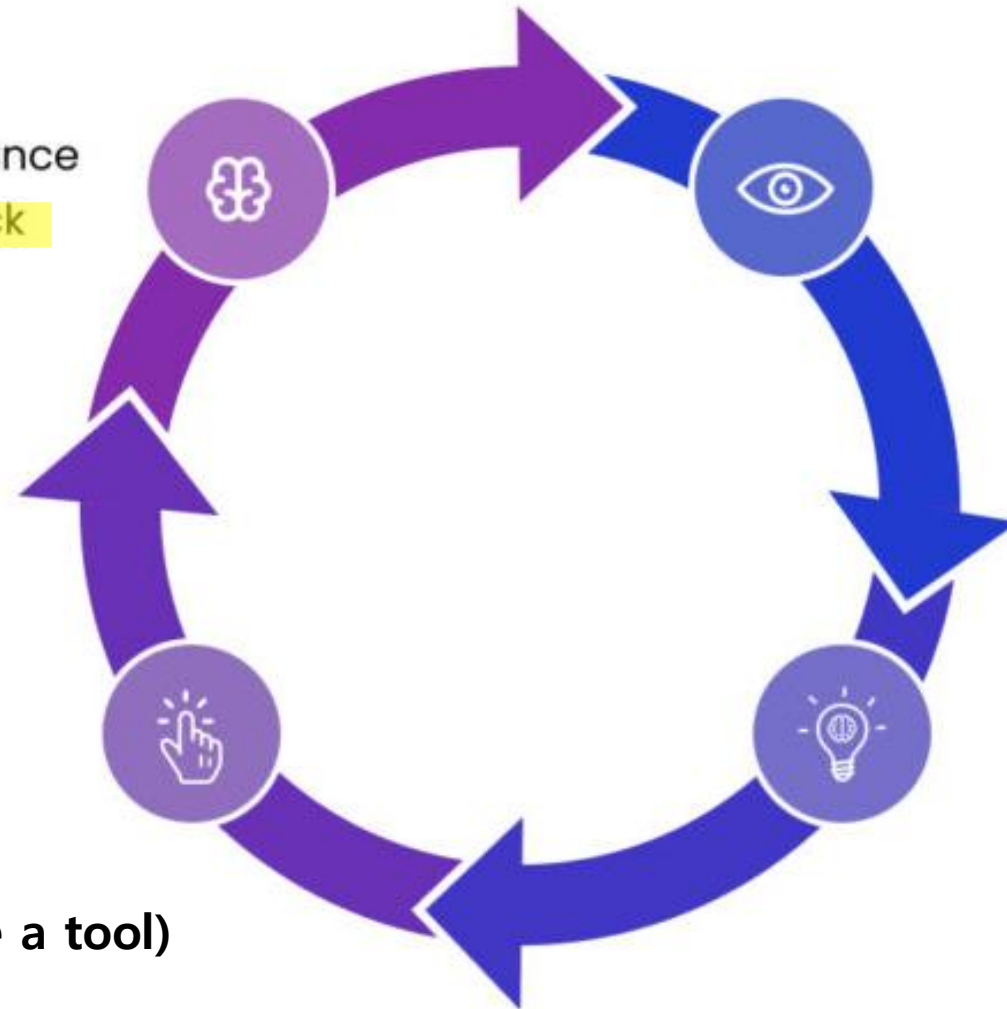
Think

Interpret data and
select action

※ Decide what to do next

Act

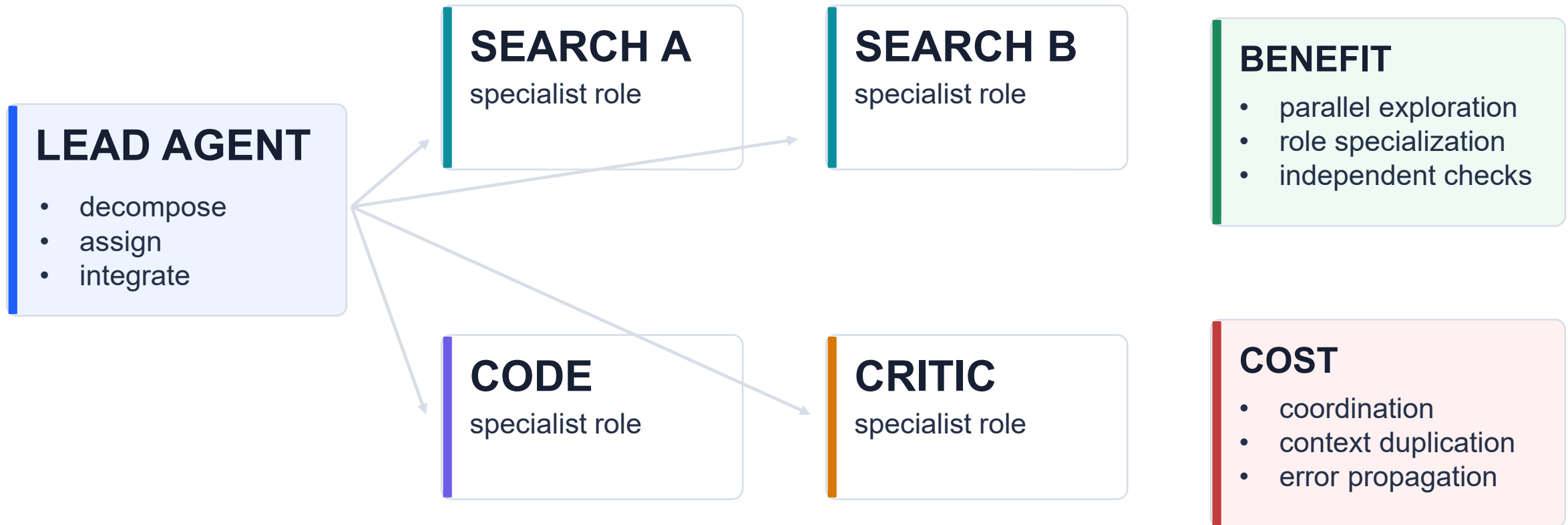
Execute action
in environment



(call an API, generate text, use a tool)

Why Multiple Agents? For complex task

Parallelism and independent checking can help — but coordination is not free.



Parallel exploration and specialization

Agent Failure Modes Are System Failure Modes

Planning drift

The agent loses the original objective over many steps.

Compounding errors

Small mistakes become irreversible downstream.

Tool misuse

Wrong command, API parameters, file, or UI target.

Cost explosion

Loops and retries consume time and tokens.

Security

Prompt injection, data exfiltration, unsafe actions.

Evaluation

Task success is harder to measure than next-token accuracy.

Production agents need permissions, sandboxes, observable state, stop conditions, and deterministic checks when possible.

05

Building Interactive Intelligence

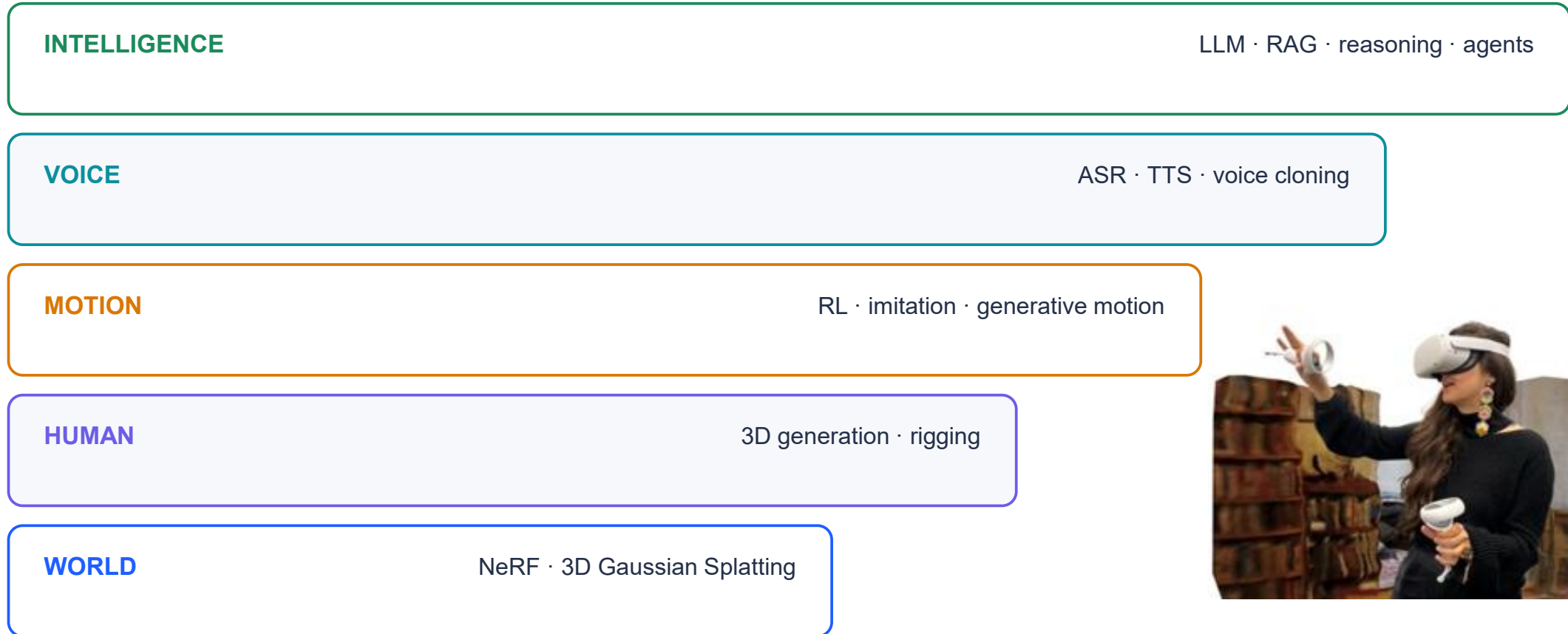
Question?

What components do we need if we want to place an intelligent AI in interactive 3D worlds?

Deep Learning Application in 2026

The Interactive Intelligence Stack

The integration — not an isolated model — is the final artifact.



TERM PROJECT · Combine multiple capabilities into one working XR interactive system.

From Pixels to Persistent 3D Worlds using SPATIAL AI

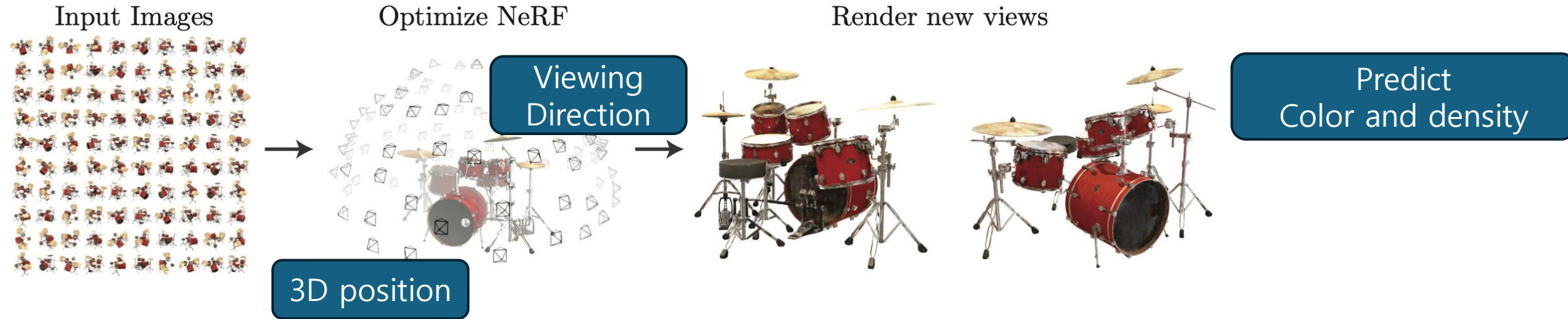


The overall trend and direction

(1) scene-specific reconstruction → (2) generalizable prediction
→ (3) dynamic and physically useful scene representations

NeRF: A Neural Network Represents the Scene

Week 5 question: How can a neural network generate a view it has never seen?



NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall et al. 2020, ECCV

**Learn a continuous radiance field,
then volume-render novel views**

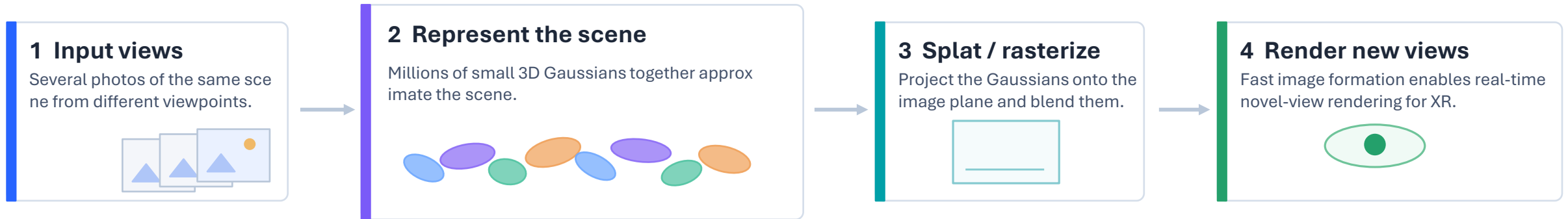


<https://arxiv.org/pdf/2003.08934>

<https://github.com/yenchenlin/nerf-pytorch>

3D Gaussian Splatting: A Scene Made of Explicit 3D Gaussians

Unlike NeRF, the scene is stored as many visible 3D primitives that can be rendered very quickly.



Each Gaussian stores

μ position

Σ scale + rotation

α opacity

C color

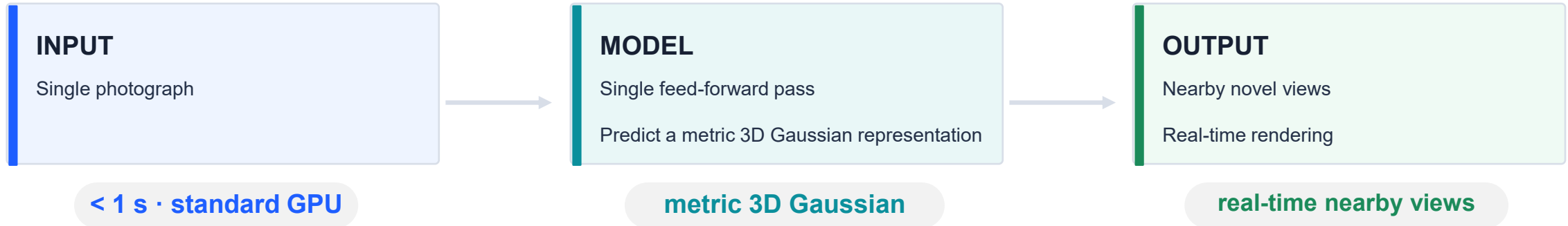
2026 direction:
Predict Gaussians directly;
add semantics, dynamics, and physical attributes.

Jiang et al. 2024 VR-GS: A Physical Dynamics-Aware Interactive Gaussian Splatting System in Virtual Reality



SOTA Snapshot · SHARP

Apple ML Research · presented at ICLR 2026



Pedagogical point: the model predicts a renderable 3D scene representation directly from a single image.



2026 Spatial AI: Reconstruction Becomes Unified

❖ iLRM · CVPR 2026

- Iterative Large 3D Reconstruction Model (iLRM)
- Feed-forward & scalable 3D Gaussian reconstruction.

3D reconstruction is becoming
more generalizable and scalable

❖ UNIFIED 3D + SEMANTICS

- Generalizable scene representations increasingly combine geometry with open-vocabulary semantics.

Including semantics like
“what the objects are?”

❖ D4RT · 2026

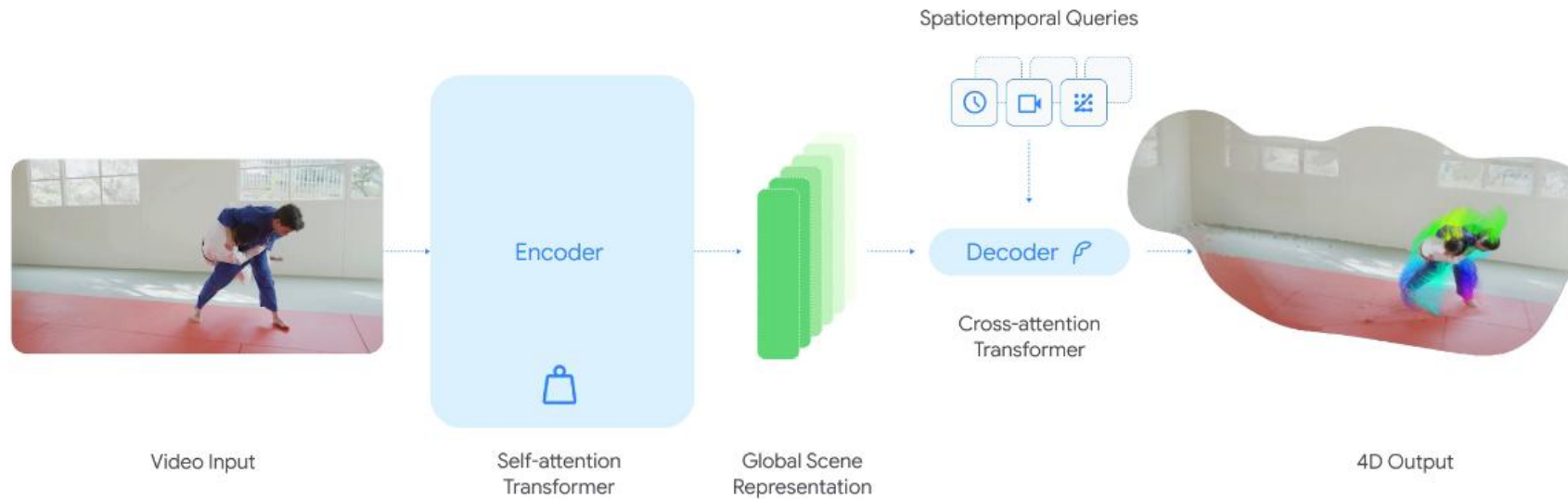
- One framework for dynamic 4D reconstruction + tracking across space and time.

4D reconstructions include
“Time and tracking”

**Reconstruction is shifting from “make a pretty view”
Toward persistent 3D perception for interactive systems.**

4D Reconstruction = 3D + Time with Tracking

Google DeepMind / D4RT: Teaching AI to see the world in four dimensions (2026)



Reconstruct geometry and track motion across space and time in one model.

<https://deepmind.google/blog/d4rt-teaching-ai-to-see-the-world-in-four-dimensions/>

06

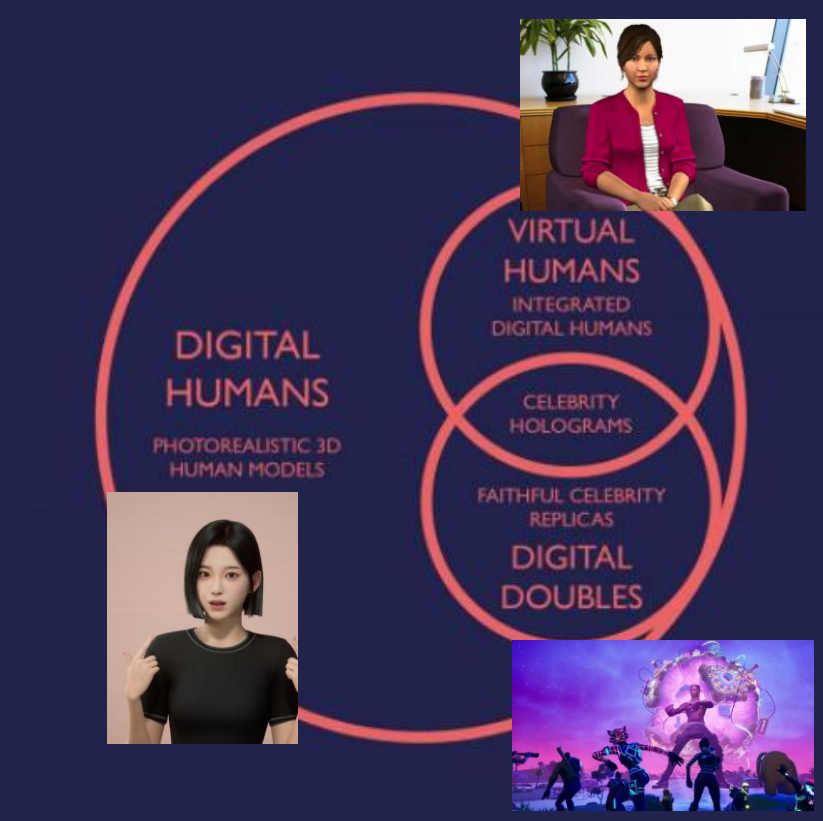
Digital Humans & Embodied Motion

A 3D human is not just geometry —

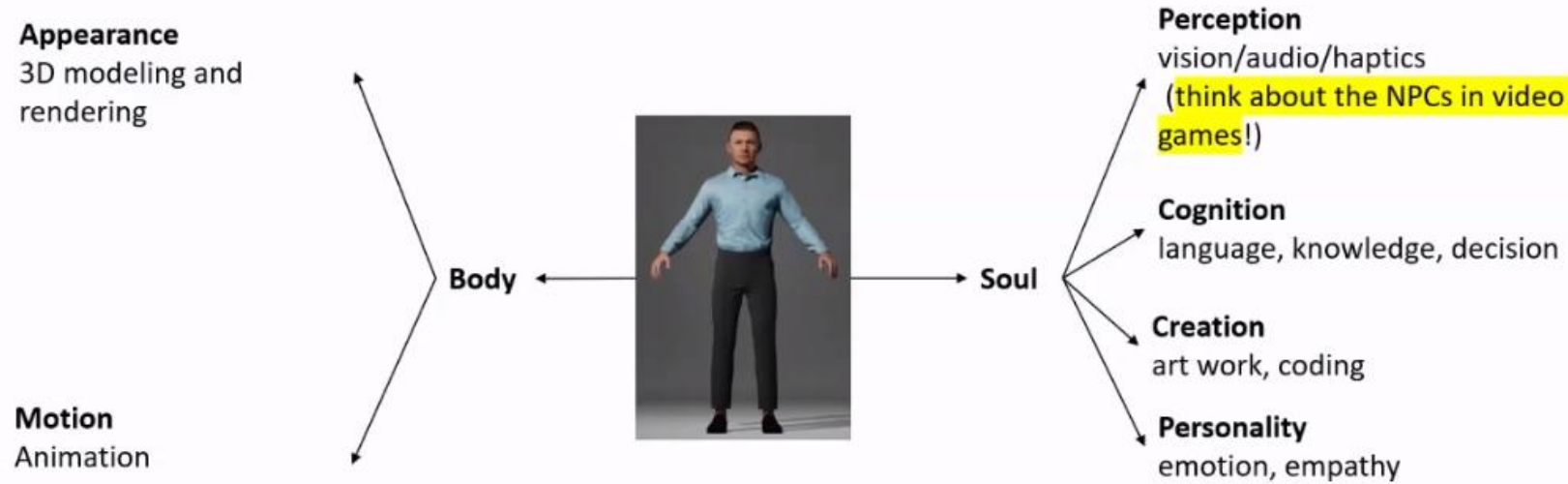
To make a digital human interactive,
it must be rigged, animated, controlled, and eventually grounded in physics.

Deep Learning Application in 2026

What Makes a Digital Human Interactive?



Digital Human = Body + Soul



DNN & large generative models are super powerful, but we're in **the driver's seat**

2026 Direction: From Static Reconstruction to Animatable Avatars

The research target increasingly includes pose, clothing, motion-dependent appearance, and fast personalization.

INPUT

One or a few images

Possibly sparse motion / pose cues



GENERATIVE / RECONSTRUCTION PRIOR

Learned 3D human prior

Geometry + appearance + articulation



ANIMATABLE OUTPUT

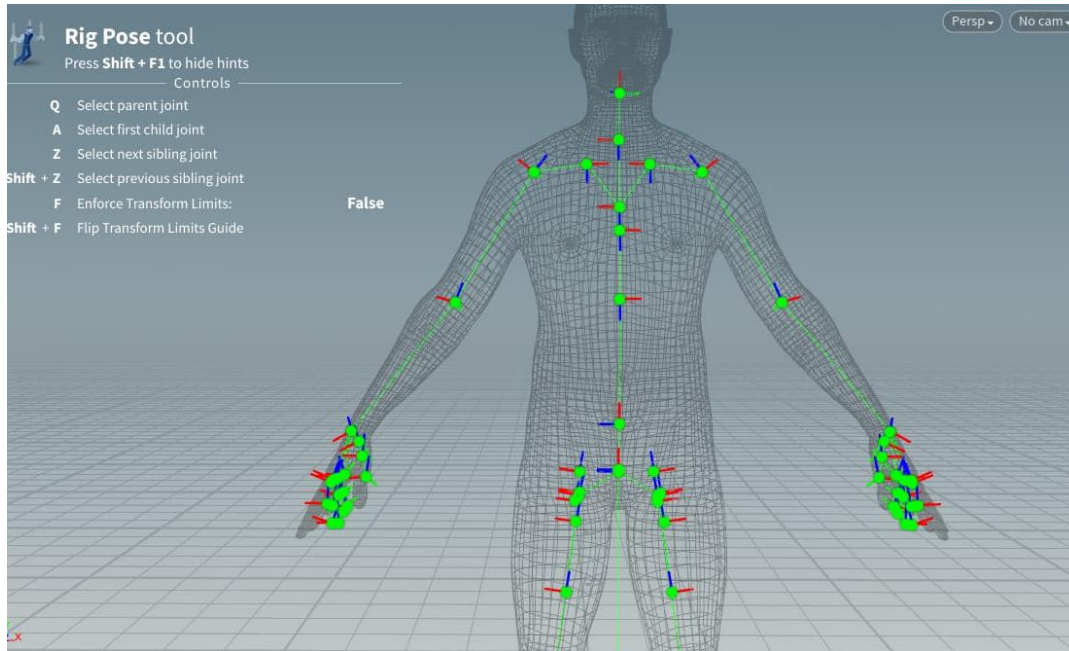
Avatar that can be posed / driven

Increasingly models cloth and motion-dependent effects

Examples in 2026 research include zero-shot animatable-human reconstruction and large-scale avatar pretraining.

Rigging Turns Geometry into a Controllable Body

Week 7 · RigNet lineage



MESH / BODY
geometry

SKELETON + SKIN WEIGHTS
joints · hierarchy ·
deformation

POSE
control variables

ANIMATION
motion

A representation becomes useful for interaction when it exposes control variables.

This principle reappears in world models and robotics: representations matter because agents must act through them.

DeepMimic: From Playing Back Motion to Learning How to Act

How do we generate motion?

Traditional Way: ANIMATION PIPELINE

Motion capture



retarget



playback

The motion is authored / recorded first.

POLICY LEARNING

Reference motion + physics simulation

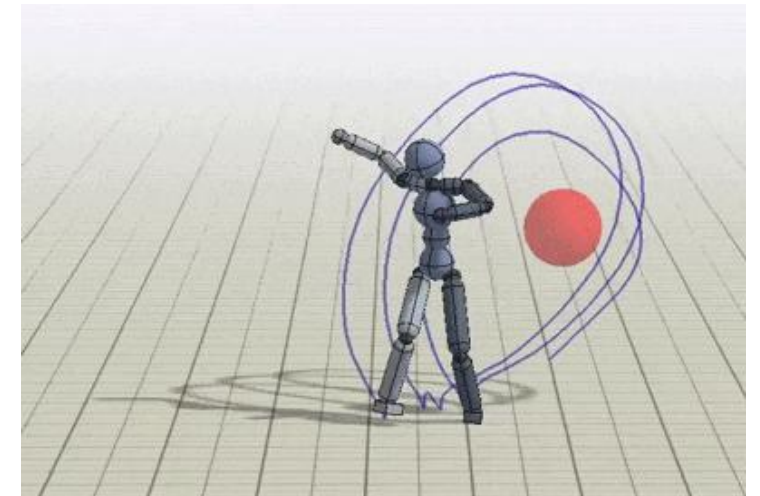
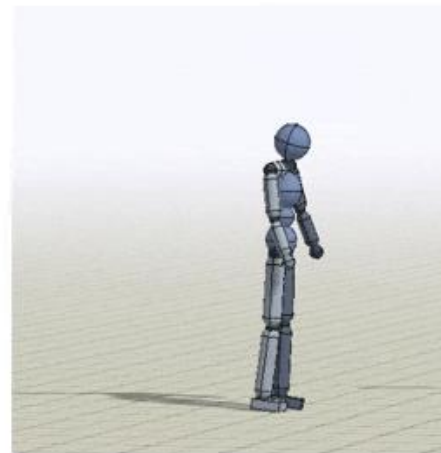
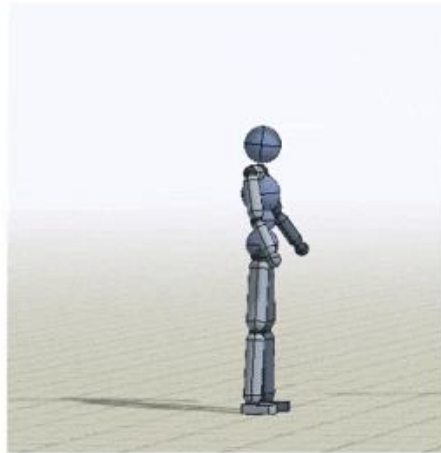
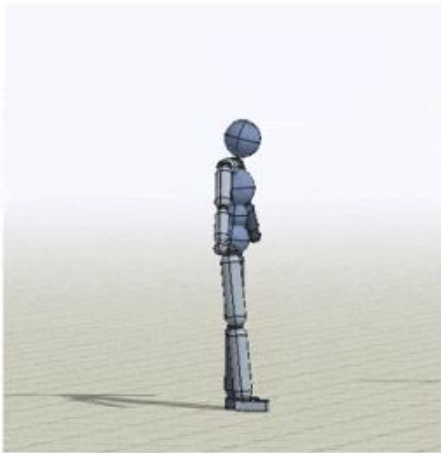


Policy $\pi(a | s)$



Physically simulated character

The controller learns actions.



2018. DeepMimic: Example-Guided Deep Reinforcement Learning of Physics-Based Character Skills, SIGGRAPH

2026 Motion Generation: Real-Time + Controllable

NVIDIA ARDY · SIGGRAPH 2026

<https://research.nvidia.com/labs/sil/projects/ardy/>

INPUT CONTROL

- Online text prompts
- root paths / waypoints
- full-body keyframes
- sparse joint constraints

ARDY

- Autoregressive diffusion
- streaming generation
- long-horizon constraints
- real-time responsiveness

OUTPUT / USE

- Interactive human motion
- animation / simulation
- can bridge to humanoid control with a tracking controller

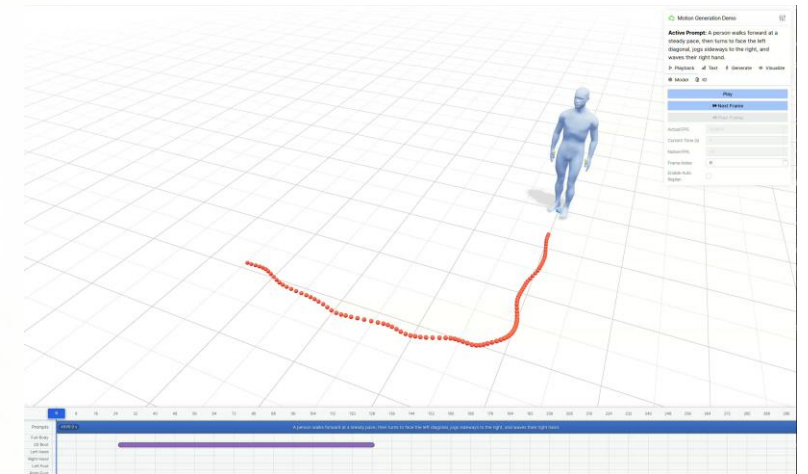


2018 Siggraph AISA

Neural State Machine for Character-Scene Interactions



Application: Scene Traversal



Research direction: combine generative motion priors with responsive control loops.

07

Speech & Voice

From ASR/TTS pipelines to native speech-to-speech interaction and controllable voices.

Deep Learning in 2026

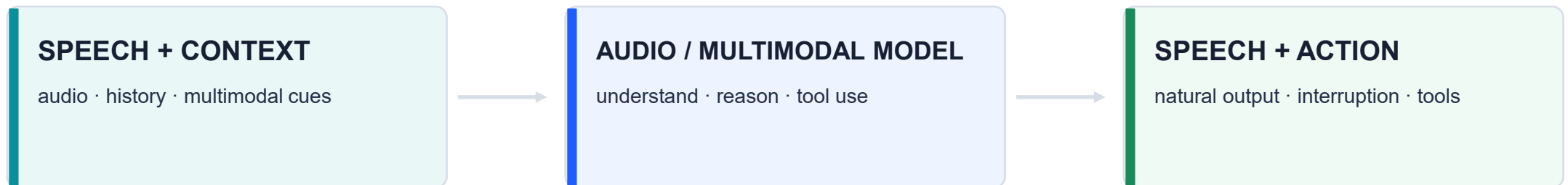
Speech AI: Pipeline → Native Conversation

Course Weeks 11–12

CLASSICAL PIPELINE



NATIVE SPEECH MODEL



Latency, prosody, turn-taking, and interruption handling become part of interaction design.

2026 SOTA Direction: Controllable Zero-Shot Speech

DISENTANGLE

- Content
- Prosody
- Timbre

Separate what is said, how it is said, and who so sounds like the speaker.

MODEL

Codec tokens + language model

Prosody continuation from a style prompt

Target timbre injected at decoding

AUTHOR-REPORTED RESULT

Competitive voice cloning

Superior zero-shot prosody control

Audio samples publicly available

**Voice synthesis is no longer only “copy a voice”;
controllability and speaker consent become system-level design issues.**

2026: Speech Models Become Interactive Agents

Voice is moving from an I/O wrapper around text toward a first-class interaction modality.

LISTEN

low-latency streaming
audio

UNDERSTAND

context + multimodal
reasoning

ACT

tools / external systems

SPEAK

natural, controllable
output

**Speech can be fused with gaze, head / hand pose,
scene context, and spatial audio inside the same agent loop.**

08

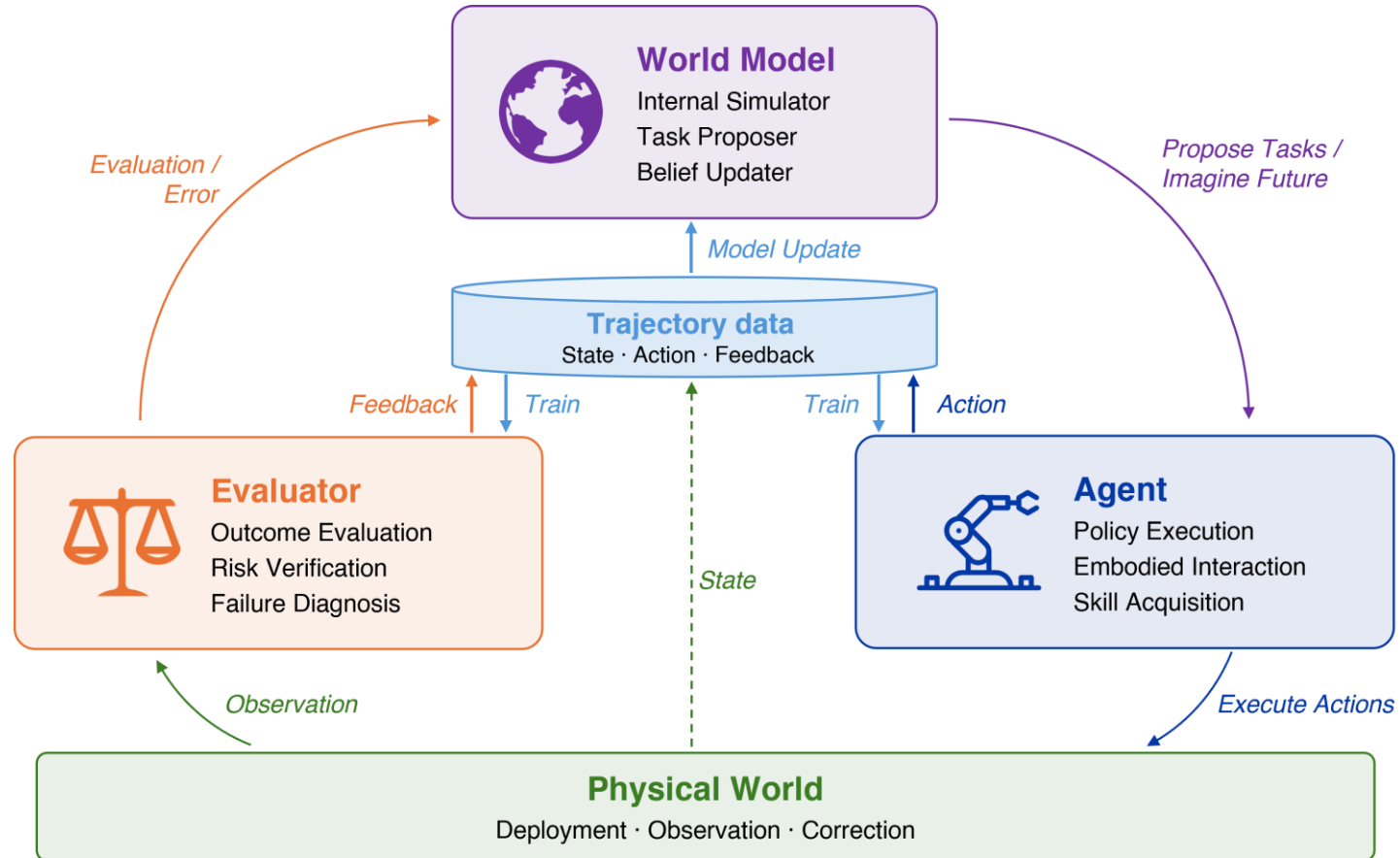
World Models & Physical AI

Generation becomes action-conditioned simulation; agents must predict what the world will do next.

Deep Learning in 2026

What Is a World Model?

A useful minimal definition: predict how the world changes when actions are taken.



PERCEIVE → PREDICT → PLAN → ACT → OBSERVE → repeat

Video Generator vs. World Model

The boundary is getting blurry; action-conditioned closed-loop behavior is the key distinction.

VIDEO GENERATOR

Input prompt / image
↓
Generate a sequence
↓
Watch the result

Primary question: "Does the sequence look coherent?"

WORLD MODEL

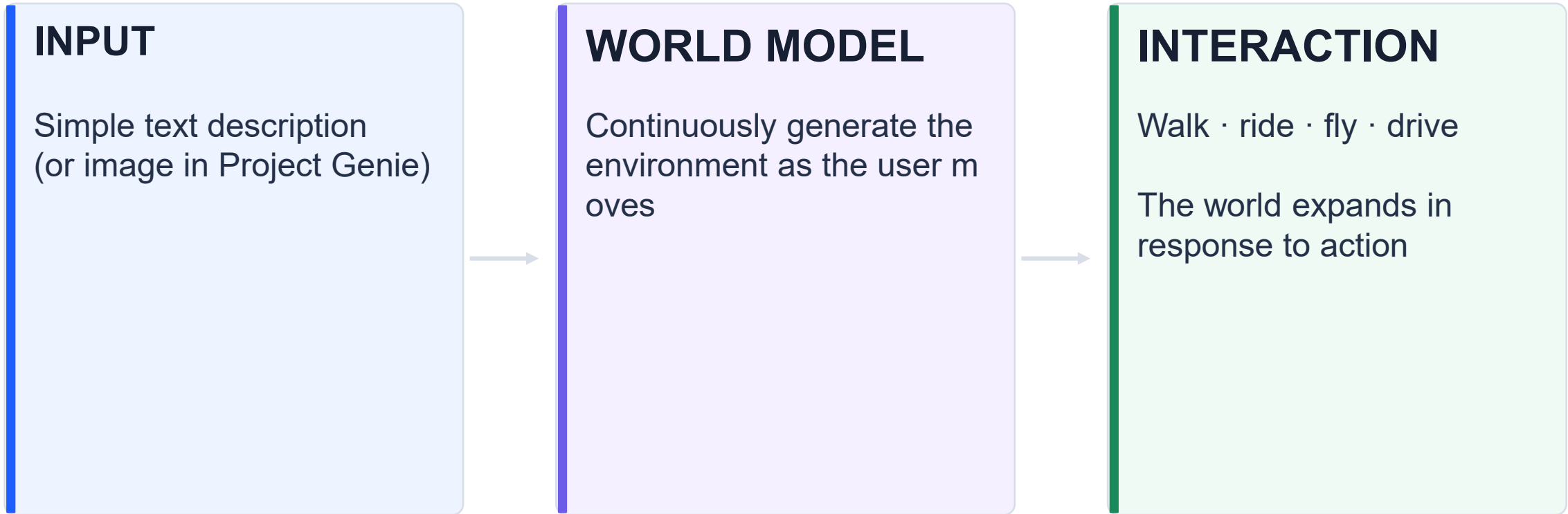
Current state + user / agent action
↓
Generate / predict next state
↓
Interact and continue

Primary question: "Does the world respond coherently to action over time?"

2026 extension: NVIDIA Cosmos 3 explores omnimodal world models spanning language, image, video, audio, and actions for Physical AI.

Genie 3: Generate a World You Can Explore

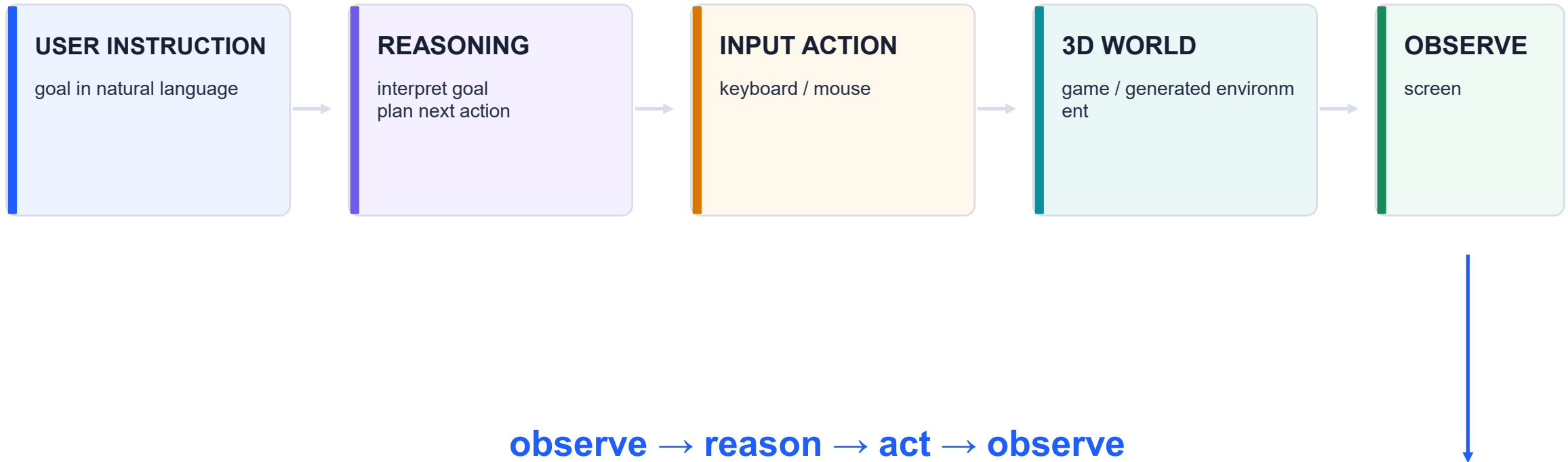
Google DeepMind · general-purpose interactive world model



A world model produces an interactive trajectory, not just a pre-rendered clip.

SIMA 2: Put an Agent Inside the World

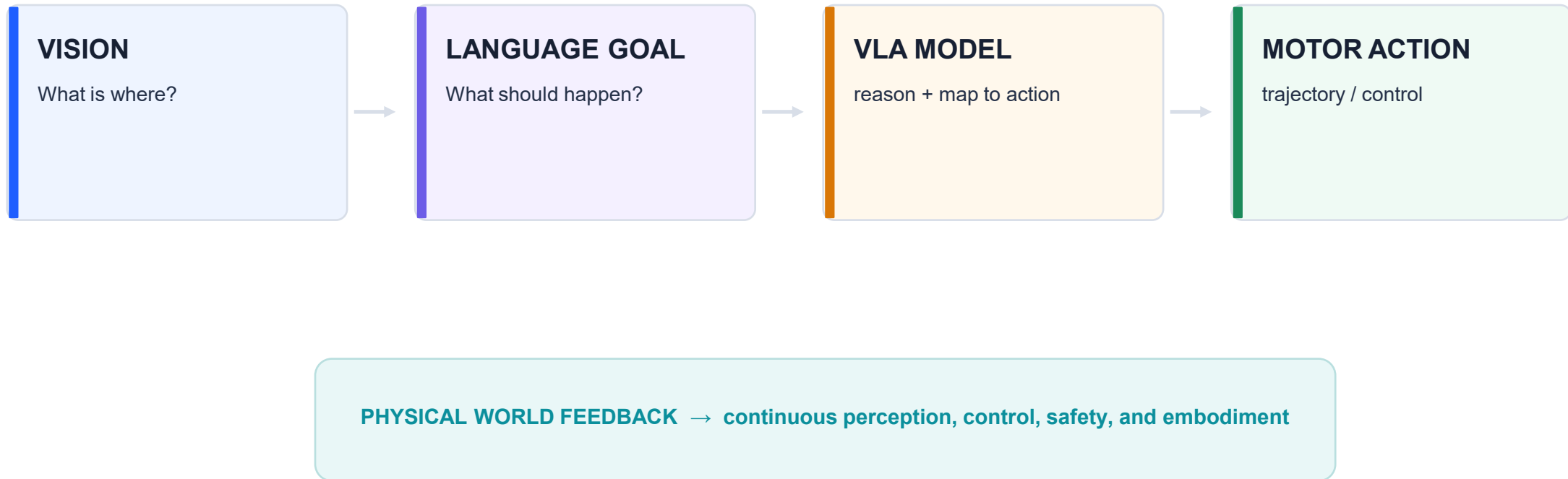
A clean bridge from agentic AI to XR-like 3D environments.



Research direction: generalist agents act through visual observations and ordinary controls, including in previously unseen 3D worlds.

Vision–Language–Action (VLA)

The “output token” can encode an action trajectory or control command — not only text.



Gemini Robotics 2 is a 2026 example of whole-body VLA control; ER 2 adds high-level embodied reasoning and orchestration.

Summary of DL trends

1 MULTIMODAL

AI is expanding from text and images to audio, video, 3D, and action.

2 DELIBERATIVE

Inference is becoming a process: reason, search, verify, use tools, retry.

3 INTERACTIVE

The frontier is shifting from producing answers to building systems that act in digital and physical worlds.

Model quality still matters. But in 2026, system design matters just as much.

Return to the Opening Question

Why can an AI solve an Olympiad problem
— but still struggle to clean a room?

Because

- physical world is noisy and uncertain.
- Objects follow continuous physics.
- Actions must happen in real time.
- Small errors accumulate over long sequences.
- Mistakes have physical consequences.
- And every home is different.

A digital reasoning system can often retry.



A physical agent has to survive the consequences of its actions.

- ➔ Strong performance on a digital benchmark does not automatically mean robust embodied intelligence.
- ➔ Gap between digital intelligence and embodied intelligence is the most important research frontiers.